

Profile Summary

Senior Software Engineer with 5+ years of experience building Linux-based networking, security, telecom and embedded systems solutions. Expertise includes C/C++, VPN technologies, DPI, Zero Trust architectures, 5G network software, and performance-critical distributed systems. Experienced in delivering production-grade platforms spanning Linux kernel networking, cloud-native deployments, and carrier-grade telecom infrastructure.

Name	Mr. Neeraj Chaurasiya
Qualification	B. Tech (Computer Science) from Amity University.
Total years of experience	5 years 11 months (Jul 2020 – Present)

Skills:

Technology / Domain	5G Network Software, PHY Emulator (PhyEm), SMF Node Development, XRUM, VPN Technologies (IPSec, SSL VPN), Zero Trust Architecture (ZTNA), Deep Packet Inspection (DPI), Network & Cyber Security, Distributed Systems, System Design (HLD/LLD), Cloud Platforms (AWS, GCP), Generative AI & Agentic AI system
Operating Systems/Platforms	CentOS 8.4, Ubuntu 20.04/22.04/24.04, Yocto (Custom OS), RHEL 9.8, VS Code, MobaXterm, XShell
Programming / Scripting Language	C, C++, Python, Shell Scripting
S/W Engineering Methodologies / Tools	MS Visual Studio, Perforce, Git, Jenkins, Wireshark, Docker, Kubernetes, TeamCity, Jira, Bitbucket, Confluence

Employment History (in reverse chronological order including your current job):

Name of Organization	Last Held Designation	From	To
Hughes Systique Corporation	Senior Software Engineer	03/2023	Till Now
Capgemini	Engineer	07/2022	03/2023
Reliance Jio Info Pvt Ltd	Deputy Manager	07/2020	07/2022

Project Details (in reverse chronological order):

Project Name	5G Network (Research & Development)		
Project Description	R&D role at Reliance Jio focused on 5G SMF (Session Management Function) development. Built back-to-back SIP Application Server framework and ASN Encoder/Decoder in C/C++ on Linux. Owned full development lifecycle: requirements, design, coding, unit testing, load testing, debugging, and release/hotfix management.		
Duration	From: 07/2020	To: 07/2022	
Tools	Visual studio, Git,		
Language	C++, Linux		
Role	Engineer		
Responsibilities	• Captured SMF requirements and drove end-to-end development: design, coding, unit testing, debugging, and load testing of 5G core application modules. • Built and validated SIP Application Server back-to-back framework and ASN Encoder/Decoder in C/C++ on Linux. Managed documentation, release cycles, and hotfixes across 5G feature deliveries.		

Project Name	5G OAM(XRUM)		
Project Description	XRUM (ORUM) is a C++ software block on Samsung Distributed Unit (DU) for 5G OAM. It manages and configures xRAN/O-RAN compliant Remote Units (RUs) via Netconf connections using the Libncc library, handling RU configuration, alarms, and performance stats. Subscribes to the Configuration Database (CDB) for dynamic RU config updates triggered by the CU (Centralized Unit).		
Duration	From: 08/2022	To: 02/2023	
Tools	Mobaxterm, Visual studio, perforce, Jenkins		
Language	C++, Linux		
Role	Engineer		
Responsibilities	Developed and tested the XRUM (ORUM) block on Samsung 5G DU for xRAN/O-RAN Remote Unit management. Implemented Netconf-based RU lifecycle management (configuration, alarms, performance stats) via Libncc, and integrated CDB subscription for real-time RU config change processing triggered by the CU.		

Project Name	PhyEmulator (PhyEm) for 5G Networks		
Project Description	5G PHY-layer emulation component enabling DU-UE communication via FAPI/nFAPI interfaces. Manages cell, carrier, and UE configurations with SFN/slot-based timing synchronization. Processes DL/UL PHY messages (data + control signaling) for interoperability testing and validation of 5G network functionality.		
Duration	From: 03/2023	To: 10/2024	
Platform/Tools	C++, Visual Studio, MobaXterm, Git, Bitbucket, JIRA, Confluence, SCons, TeamCity, CppUTest, Code Collaborator, Intel Inspector, BoundsChecker, Lucidchart, draw.io		
Language	C/C++, Python		
Role	Engineer		
Responsibilities	- Designed and implemented PHY-layer emulation (FAPI/nFAPI) supporting DL/UL message processing for up to 8 concurrent UEs across multiple cell and carrier configurations. - Maintained SFN/slot-based timing synchronization at every slot boundary, ensuring accurate DL_TTI.req/TX_DATA.req forwarding between DU and UE higher layers. - Automated unit testing with CppUTest; built and managed C++ codebase via SCons with CI/CD on TeamCity, reducing integration failures across release cycles. - Debugged and resolved memory leaks and corruption using Intel Inspector and BoundsChecker, improving stability in long-duration PHY emulation test runs. - Conducted code reviews via Code Collaborator and Bitbucket PRs; tracked issues and documentation in JIRA and Confluence across the full project lifecycle.		

Project Name	QZTNA — Zero Trust Network Access Solution
Project Description	Real-time Zero Trust Network Access solution integrating nProbe (IPS mode) with a quantum-safe VPN (QVPN) on RHEL 9.5. Enforces L7 protocol, hostname, and GeoIP-based per-user traffic policies via a ZeroMQ pub-sub Policy Engine backed by the Maestro management DB. Uses iptables/NFQUEUE to steer VPN tunnel traffic to nProbe for DPI, with dynamic CN-to-IP mapping for per-user enforcement. Integrates Elasticsearch/Kibana for flow visibility and a React-based

	Maestro GUI for policy lifecycle management. Performance tested at 941 Mbits/sec baseline; nProbe inline overhead only ~6.6% (879 Mbits/sec).
Duration	From: 10/2024 To: 07/2025
Tools	DPI Tool (IPS mode), ZeroMQ (ZMQ), iptables, NFQUEUE, Elasticsearch, Kibana, Wireshark, tcpdump, iperf, Flask (REST API), React (Maestro GUI)
Language	C/C++, Python
Role	Senior Software Engineer
Responsibilities	Led end-to-end delivery of a DPI-based traffic inspection and policy enforcement solution, including requirements analysis, architecture design, development, testing, deployment, and production support; designed a modular and scalable architecture comprising nProbe Manager, Policy Engine, Policy Reader, Policy Enforcer, REST Server, and Maestro GUI integration; conducted PoCs and comparative evaluations of Suricata, custom nDPI-based implementations, and nProbe IPS to assess detection accuracy, performance, scalability, and integration feasibility before selecting nProbe IPS as the DPI engine; and designed multi-client test topologies to validate end-to-end traffic classification, policy enforcement, and VPN tunnel isolation. visited customer sites to perform solution installation, deployment, configuration, troubleshooting, and production rollout support.

Project Name	QConnect Gateway Client Box (QBox)
Project Description	Linux-based quantum-safe VPN gateway appliance with full network stack: real-time DPI on LAN/WAN via nProbe, Flask/MongoDB policy API with pyroute2 interface control, Dnsmasq DNS/DHCP, PPPoE server (rp-pppoe + CHAP), WPA2-Enterprise Wi-Fi (Hostapd + FreeRADIUS/802.1X), and iptables NAT with systemd-managed startup. Integrated with Maestro control plane for centralized, zero-touch policy management.
Duration	From: 07/25 To: 10/25
Tools	nProbe, Dnsmasq, Hostapd, FreeRADIUS, rp-pppoe (PPPoE), iptables, pyroute2, Flask, MongoDB, systemd, QVPN RAC Client
Language	Python 3.10, Linux (Ubuntu), C/C++
Role	Senior Software Engineer
Responsibilities	Designed and implemented the full network stack of the QBox

	gateway: DPI, DHCP/DNS, PPPoE, WPA2-Enterprise Wi-Fi, and NAT routing. Integrated the device with the Maestro control plane for centralized policy management and monitoring. Validated end-to-end connectivity, authentication flows, and policy enforcement across LAN and WAN interfaces.
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Project Name	SMOAD IPsec VPN Gateway with Quantum Entropy Integration
Project Description	Deployed StrongSwan 6.0.3 from source on Ubuntu 22.04 with IKEv2/IPsec full-tunnel VPN, enabling 30+ build flags (EAP-TLS, EAP-RADIUS, DHCP, Curve25519/448, DRBG). Built a custom CA and per-client .p12 PKI for certificate-based mutual auth on Android. Integrated QDice Quantum Entropy (QEaaS) via a custom C plugin (qdice_rng_plugin.c) with ring-buffer feeder. Deployed StrongMan dashboard (Django/Gunicorn/Nginx) and a Prometheus-compatible exporter for real-time IKE/SA stats, quantum entropy health, and attestation trust metrics. Configured iptables NAT and firewall rules for the IPsec subnet.
Duration	From: 10/2025 To: 02/2026
Tools	StrongSwan 6.0.3, swanctl, charon-systemd, iptables, nftables, Prometheus, Django, Gunicorn, Nginx, OpenSSL, Android StrongSwan Client
Language	C, Python, Linux (Ubuntu 22.04)
Role	Engineer
Responsibilities	Designed and implemented the full IKEv2/IPsec VPN stack from source compilation to production deployment. Built and integrated a custom quantum entropy plugin to enhance VPN cryptographic key generation security. Set up monitoring and observability infrastructure (Prometheus exporter, StrongMan dashboard) for operations teams. Validated mutual certificate-based authentication and end-to-end VPN tunnel functionality on Android clients.

Project Name	WSR AI Agent - AI-Powered Project Monitoring Platform
Project Description	Built an AI agent for automated Jira project monitoring, sprint analysis, and stakeholder reporting. Bypassed Jira API limitations using a browser-extension and cookie/session-based authenticated data collection mechanism. Developed an advanced analytics dashboard with project health, productivity, and risk metrics. Automated

	generation of AI-powered email summaries highlighting progress, blockers, and actionable insights.		
Duration	From: 02/26	To: 04/26	
Platform/Tools	Python, Jira, Browser Extensions, AI/ML APIs, Analytics Dashboards		
Language	Python, JavaScript		
Role	Engineer		
Responsibilities	- Developed AI agent for automated Jira monitoring and sprint analysis. - Bypassed API limits via browser-extension and session-based data collection. - Built analytics dashboard for project health, productivity, and risk metrics. - Automated AI-powered email summaries for progress, blockers, and insights.		

Project Name	Princess Cruise Line
Project Description	Production deployment and embedded systems engineering across a ship cruise fleet. Managed OTA rollout of LokAgent, DP-Monitor, and DoorPanel packages fleet-wide, resolving upgrade failures on non-compliant devices. Built a Yocto (ARMv7) embedded Linux environment from scratch for shipboard door panel hardware. Designed CI/CD pipelines for Panel-UI and Panel-App; built and validated the MID SDK across Android, Web, and Unity targets. Developed and automated startup package generation workflows for cruise deployments, and designed device rules and version-based configuration mechanisms to enable mass deployment, automated provisioning, configuration management, and scalable software rollout across large fleets of devices.
Duration	April 2026 – Till now
Tools	Yocto Project, Docker, CI/CD Pipelines, MID SDK (Android/Web/Unity), systemd, XS Device Platform, OTA Deployment
Language	Python, Bash, Linux (Ubuntu 20.04), ARMv7
Role	Senior Software Engineer
Responsibilities	Led OTA deployment and upgrade management across a 17-ship cruise fleet. Built and maintained embedded Linux (Yocto) build systems and

	CI/CD pipelines. Diagnosed and resolved production defects via log analysis and targeted package upgrades. Maintained consistent software versions fleet-wide through controlled SU-level upgrades.
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